

Program : Diploma in Wood and Paper Technology	
Course Code : 4122	Course Title: Paper Technology
Semester : 4	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- To introduce the students with the concepts of stock preparation and sheet forming processes of papermaking.
- To provide knowledge on design, operation and maintenance aspects of paper machine.
- To familiarize students with recovery of energy and chemicals used in pulping processes with due techno-economic and environmental considerations.
- To make the students conversant about various environmental issues and management aspects related to Paper Industries.

Course Prerequisites:

Topic	Course code	Course name	Semester
Chemistry of wood, colloids		Wood chemistry	3
Pulping		Pulp technology	3
Pollution control techniques		Environmental science	2

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Summarize the processes and operations in papermaking	16	Understanding
CO2	Outline the constructional details, operation and maintenance of paper machine	13	Understanding

CO3	Describe coating and finishing operations and the properties of paper	16	Understanding
CO4	Apply the chemical recovery operations, effluent treatment and pollution control techniques	13	Applying
	Series Test	2	

CO - PO Mapping:

Course Outcomes	PO1	PO 2	PO3	PO4	PO5	PO6	PO7
CO1							
CO2	3						
CO3	3						
CO4	3				3		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Summarize the processes and operations in papermaking		
M1.01	Explain the Beating and Refining of Pulp	4	Understanding
M1.02	Discuss Storage Chest and Agitators in Stock Preparation Side	4	Understanding
M1.03	Describe fibre recovery	2	Understanding
M1.04	Explain Sizing, Filling, Dyeing and additives used	3	Understanding
M1.05	Comprehend the screening, cleaning and deaeration	3	Understanding

Contents:

Stock preparation- Defibration - Hydra Pulper - , Kneaders, Deflakers ; Storage chests and agitators - different types of stock and machine chests - agitator assemblies ; Beating & Refining processes - Theory - factors influencing beating and refining - freeness - Hydration - Fibrillation - and their effects on paper properties ,different types of beaters & refiners - Hollander beater- conical refiner, narrow wide angle shallow angle refiners single and double disc, Types of consistency controllers - working principles and stock proportioning

Fibre recovery - method , types of save all - settling and sedimentation. Conical saveall, Flotation saveall, SP Krofta, Filtration- need ,stuff box

Purpose of sizing - internal and surface sizing - sizing chemicals such as rosin, alum, and wax their preparation, - parameters influencing sizing , acidic ,alkaline and neutral Sizing
 Surface sizing - Sizing chemicals starch as a sizing agent Size presses - straight horizontal, inclined and speed sizes

pH, order of addition - pulp characteristics ;Fillers -Types of fillers, properties of fillers, effect of fillers on optical and mechanical properties of paper - theory of retention; Dyes and pigments, dying of paper ;Control Chemicals at the Wet End: Retention aids, drainage aids, defoamers, deflocculants and pitch controlling agents. Strength Additives: Dry and wet strength additives, mechanisms of strength development, factors affecting wet and dry strength properties

Cleaning and screening - settling, centrifugal screens - HD and LD cleaning - Different methods of cleaning - pressure screens - centrifugal & cleaners - sorters Deaeration - purpose - common type of dearators - decculators.

CO2	Outline the constructional details and operation of paper machine		
M2.01	Illustrate Paper Machine Head box	4	Understanding
M2.02	Comprehend Paper Machine Wire part	4	Understanding
M2.03	Identify the Press part and Dryer part of Paper machine	5	Understanding
	Series Test - I	1	

Contents:

Head box - head box and its parts - distribution section, middle section- slice straight, convergent and w type of head box Type of head boxes- open and closed - role of distributor roll - high turbulence head box - different type of manifolds - head box over Flow, head box re circulation, jet wire ratio, head calculations (simple problems only) Wire part - General arrangement and description of paper machine system and sheet formation type of paper machine wire part - cylinder mould- fourdrainer - twin wire formers - special former combinations - different components of wire part foaming medium wire weave pattern sheet transfer factors affecting drainage and sheet formation

Wet Pressing: Press section, mechanism of pressing, types of presses, operating variables, press felt characteristics

Drying - principle, pre drying factors- factors affecting drying - heat content temperature moisture different types of dryers and its arrangement yankee cylinder (MG). Multi cylinder (MF) combined - infrared radiator, flaket use of air in paper drying Steam and condensate removal - cascade - pressure regulated differential pressure regulated -- ejectors - Thermo compressors- Condensate removal method - bucket, scoop siphon, rotating and stationary. Rimming calculation of heat transfer, heat transfer rate Overall heat transfer coefficient Dryer hoods open - closed high velocity exhaust fans, dryer felts and screens - its cleaning and drying permeability's Common troubles drying - psychrometry.

CO3	Describe coating and finishing operations and the properties of paper		
M3.01	Discuss the coating of paper	4	Understanding

M3.02	Classify Finishing operations	4	Understanding
M3.03	Explain speciality Paper and types of paper	4	Understanding
M3.04	Summarize the properties of paper	4	Understanding

Contents:

Coating, calendaring & reeling - coating pigments - binders - coating formulations - coating equipment's, air knife coater, brush coater, cast coating, flexible blade coater, gravure coating, knife coating reverse roll coater - transfer roll coater - curtain coating

Calendering - theory of calendering - effect of calendering on paper quality open side closed side calender swimming rolls - Importance of moisture on paper during calendaring common calender troubles hard and soft spots sheet blackening cuts

Finishing -- winding, different types of winder, converting, slitting, sheet cutting rotary simplex duplex guillotines ,sorting ,counting, packing -sheets - reels - roll wrapping machine weighing and labeling - scabs - reeling- pope reel - core mind reels - friction or surface drive reel

Speciality paper - converting of paper - corrugating of paper -uses of wet and dry waxing of paper laminating of paper - properties sizes etc regarding paper bags and envelopes - special characteristics of paper and board - creeping - news print , grease proof ,glassine paper and, Tissue paper bond and currency papers Kraft paper and sack draft

Basis weight, caliper, density, bulk, folding endurance, tear strength, burst strength, tensile, stretch, paper testing - sample preparations surface properties - Roughness, smoothness pick strength, abrasion resistance oil absorbency, brightness, opacity, gloss sizing degree, colour, water absorbency

CO4	Apply the chemical recovery operations, effluent treatment and pollution control techniques		
M4.01	Understand chemical and energy recovery operations in Paper Mills	5	Understanding
M4.02	Comprehend the causticizing process	4	Understanding
M4.03	Understand the effluent treatment	2	Applying
M4.04	Apply different pollution control techniques	2	Applying
	Series Test - II	1	

Contents:

Chemical recovery

Black liquor - properties - chemical composition of black liquor - desilication and oxidation
Evaporators - working principle, types of evaporators-.Steam economy calculation- recovery boiler and furnace Types - function, electrostatic, precipitator (ESP) reactions in recovery in boiler, make up chemical ID fan, FD fan, Boiler feed water properties

Causticizing Functions and chemical reaction batch and continuous process - green liquor classifiers and sludge removal Green liquor, white liquor, sludge, sulphidity, causticizing lime slakes, classifiers, white liquor classifier mud washer operation techniques - lime reburning sludge disposal - make up chemicals - properties storage, transportation methods, handling and effect of lime impurities

conventional sulphate recovery systems, Zeolite process, demineralization process, simple calculation of white liquor for cooking

Effluent treatment.- Pulp and paper mill effluent - physical and chemical nature of effluent and its treatment pollution control board and its norms chemicals used in effluent treatment such as polyelectrolyte alum etc - COD, BOD, and colour - air pollution, steam pollution, solid waste - pollution control techniques- equality control system in pulp mills

Text / Reference:

T/R	Book Title/Author
T1	Ronald.G. Macdibak, Pulp & Paper Science, Vol. I, II, III McGraw Hill Book Co., NewYork
R1	Gullichsen J. and Paulapuro H., "Papermaking Science and Technology, Book 4: Papermaking Chemistry (Ed. Neimo L.)", Finnish Paper Engineers' Association and TAPPI.
R2	Gullichsen J. and Paulapuro H., "Papermaking Science and Technology, Book 8: Papermaking Part 1, Stock Preparation and Wet End (Ed. Paulapuro H.)", Finnish Paper Engineers' Association and TAPPI.
R3	Hough G., "Chemical Recovery in Alkaline Pulping Processes" TAPPI Press.
R4	Kocurek M. J., "Pulp and Paper Manufacture, Volume 7: Paper Machine Operations (Ed. Thorp B.)", TAPPI Press

Online Resources:

Sl.No	Website Link
1	https://www.bellmer.com/paper-technology/
2	https://www.pulpandpaper-technology.com/
3	https://www.youtube.com/watch?v=I3SodZtes-c